

SCIP

Supply Chain Intelligence Platform

End User Guide — v1.0



<https://bkumars22.github.io/SupplyChainPlatformProject>

Demo Login: kumar | Password: Kumar@2026

Built by Kumara Swamy B · Staff SDET & AI Product Builder

WHAT IS SCIP?

SCIP is an AI-powered Supply Chain Intelligence Platform built to demonstrate enterprise-grade full-stack development using modern AI tools. It provides real-time supplier risk scoring, intelligent alert explanations, cost record management, and automated quality assurance — all in one platform.

Platform Highlights

	Supplier Management	Track 47+ suppliers with OTD scores, defect rates, and AI-generated risk classifications
	AI Risk Scoring	IsolationForest ML model scores each supplier — anomalies flagged automatically
	Claude AI Explanations	Natural language explanations of every supply chain alert generated by Claude AI
	Live Dashboard	Real-time KPIs — OTD averages, cost savings, active alerts, risk distribution
	Test Intelligence	51 automated Playwright tests with live pass/fail dashboard
	Cost Records	DRAFT → APPROVE workflow for supply chain cost management
	Mobile Ready	React Native mobile app on Android and iOS with shared authentication
	Enterprise Security	JWT auth, BCrypt, RBAC, OWASP Top 10, 7-layer security architecture

GETTING STARTED — 3 STEPS

Access the platform in under 60 seconds:

1

Open the Platform

Go to: <https://bkumars22.github.io/SupplyChainPlatformProject>

2

Sign In

Username: kumar | Password: Kumar@2026 | Click Sign In

3

Explore

You are now on the Dashboard. Use the left sidebar to navigate all modules.



TIP: This is a demo environment with realistic mock data. All data is pre-loaded — no setup required.

MODULE-BY-MODULE GUIDE



DASHBOARD

Your command center. Shows real-time KPIs at a glance:

- ▶ Total Suppliers — 47 active suppliers in the platform
- ▶ Active Alerts — 3 risk alerts requiring attention
- ▶ Average OTD Score — 73.4% on-time delivery across all suppliers
- ▶ Cost Savings This Quarter — \$2.4M achieved through optimization

 **TIP:** Click any KPI card to drill down into the underlying data.



SUPPLIERS

Complete supplier scorecard with AI-powered risk classification:

- ▶ View all suppliers with OTD score, defect rate, and lead time
- ▶ Risk tiers: CRITICAL (red), AT RISK (orange), SILVER (blue), GOLD (green)
- ▶ Anomaly flag — suppliers scored as anomalous by IsolationForest ML model
- ▶ Click any supplier to see full scorecard and AI risk explanation

 **TIP:** Suppliers with Anomaly = TRUE are flagged by the AI model as statistically unusual — investigate these first.



ALERTS

AI-generated supply chain alerts with Claude AI explanations:

- ▶ Each alert shows severity — CRITICAL, HIGH, or MEDIUM
- ▶ Claude AI explanation tells you exactly what the risk means in plain English
- ▶ Recommended actions included in every explanation
- ▶ Dismiss alerts once reviewed and actioned



COST RECORDS

Supply chain cost management with approval workflow:

- ▶ DRAFT — cost record created, pending review
- ▶ APPROVED — reviewed and approved by manager
- ▶ PENDING — awaiting additional information
- ▶ Filter by status, date range, and amount



TEST INTELLIGENCE DASHBOARD

Live view of all 51 automated Playwright test results:

- ▶ 51 tests across 9 modules — all passing
- ▶ Each test shows module, status, and last run time
- ▶ Auto-refreshes every 30 seconds
- ▶ This dashboard was itself tested by 10 Playwright tests

 **TIP:** This page demonstrates AI-powered test intelligence — a real-world example of QA engineering discipline applied to an AI platform.

FOR RECRUITERS & INTERVIEWERS

This platform was built independently by Kumara Swamy B — Staff SDET with 13+ years of enterprise QA experience — using Claude AI and GitHub Copilot as engineering accelerators.

What to Explore in 10 Minutes:

- 1

Login & Dashboard
See the live KPI dashboard — note the real-time data refresh and professional UI
- 2

Suppliers Page
Click on TechParts India or PrecisionMfg Chennai — both flagged as CRITICAL risk by AI
- 3

Alerts Page
Read the Claude AI explanations — natural language generated by Anthropic Claude API
- 4

Test Dashboard
See all 51 automated tests passing — including 10 tests for the AI eval pipeline
- 5

GitHub
github.com/bkumars22/SupplyChainPlatformProject — full source code

Key Technical Achievements:

Achievement	Detail
P0 Auth Bypass Found	BCrypt skipped when password null — any password accepted
51 Playwright Tests	9 modules covered including AI eval pipeline
IsolationForest ML	Unsupervised anomaly detection for supplier risk
Claude AI Integration	Natural language alert explanations via Anthropic API
LangGraph Agent	5-node StateGraph validating AI reliability end-to-end
LLM Eval Harness	Prompt consistency & hallucination rate measurement
7-Layer Security	OWASP Top 10, JWT, BCrypt, RBAC, rate limiting
Solo in 4 Weeks	Full stack — Spring Boot + React + React Native + Python AI

TECHNOLOGY STACK

Backend	Java 17 · Spring Boot 3 · JWT · BCrypt · RBAC · OWASP · 80+ REST endpoints
Frontend	React 18 · React Router · Tailwind CSS · 9 pages
Mobile	React Native + Expo · 10 screens · Android & iOS
AI Service	Python FastAPI · IsolationForest ML · Claude AI API · LangGraph
Testing	Playwright TypeScript · 51 tests · Custom reporter · CI/CD integration
LLM Eval	deepeval · Prompt consistency · Hallucination rate · Answer relevancy
DevOps	Docker Compose · GitHub Actions · Railway · GitHub Pages
Database	PostgreSQL · H2 (testing) · JPA/Hibernate
Security	Rate limiting · HTTPS · HSTS · Stack trace hiding · SQL injection audit

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